

Cooper Huntington-Bugg

Jenks, Oklahoma | cooperehuntington@gmail.com | github.com/Cooper-Bugg

PROFESSIONAL SUMMARY

Computer Science student at Oklahoma State University with a strong foundation in C++, embedded systems, and algorithmic problem solving. Experienced in developing high-performance, low-level systems and simulations with an emphasis on efficient memory and resource management. Seeking software engineering roles in C++ and embedded systems.

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, SQL, Assembly

Core Concepts: Embedded Programming, Memory Management, Bitwise Operations, Low-level I/O, Network Security

Systems & Backend: Event-Driven Architecture, Asynchronous Programming, REST APIs, Database Design

Practices: Git, Unit Testing, Debugging, Agile, CI/CD

PROJECTS

The C++ Grind — C++, HTML, CSS, JavaScript

- Built a personal C++ learning site from the ground up, designed the frontend, wired the data layer and implemented localStorage-based progress tracking, driven by a passion for learning and teaching the language to others.
- Developed 43 hand-typed C++ projects spanning the full learning arc: CLI tools, POSIX pipelines, recursive backtracking solvers, concurrency primitives (mutexes, condition variables, lock-free queues), embedded simulators and a foundational systems library built from scratch.
- Created daily interactive challenges including a C++ keyword Wordle and code challenge engine (will this compile, buffer overflow detection, output prediction) to reinforce concept retention.

Modular Discord Bot — Python, discord.py, asyncio

- Architected a modular, cog-based Discord bot with 60+ commands, separating command handlers from core logic and API integrations to support hot reloading of components at runtime.
- Developed asynchronous, event-driven systems using asyncio and aiohttp, integrating multiple external APIs while managing concurrent multiplayer state and implementing runtime monitoring features including latency tracking and error reporting.

PROFESSIONAL EXPERIENCE

Huntington ID | Web Developer

January 2020 – Present

- Maintain and update a live WordPress site, implementing SEO improvements, plugin updates, and content revisions to ensure reliability and performance.

Royale Landscaping | Landscaping Foreman

May 2016 – January 2020

- Led small teams to complete client projects on schedule, adapting plans under tight deadlines to ensure quality and client satisfaction.

ACADEMIC PROJECTS

Capstone & Software Engineering Projects — Oklahoma State University

- Built and deployed 2 full-stack applications for Card Clash and Smart Scheduler, designing and wiring both frontends, REST API integrations, and backend infrastructure using Node.js and MySQL with asynchronous logic and persistent data storage.

EDUCATION

Oklahoma State University

GPA 4.0

B.S. Computer Science | Minor in Mathematics | Expected December 2026

Undergraduate Certificate in Cybersecurity

Tulsa Community College

GPA 4.0

A.S. Computer Information Systems | A.A. Liberal Arts | 2024